POKHARA UNIVERSITY

Level: Bachelor	Semester –Spring	Year: 2020
Program: BE		Full Marks: 70
Course: Chemistry		Pass Marks: 31.5
		Time: **2** hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions:

Section - A: (5×10=50)

Q. N. 1 Arrange the following elements in the descending order (highest at the top) of its Electron Affinity value. And explain with suitable reasons for your answer 6+2+2

Carbon, Nitrogen, Oxygen and Fluorine

What are the oxidation states of Mn in KMnO4 and MnO2. Why it has different oxidation states? Explain it.

Q. N. 2 What is the principle involved in mass spectrometry? Give any two molecules which can be analysed by using mass spectrometry with explanation. 6+4

OR

Paper chromatography is an important analytical tool for a researcher. Discuss the principle involved and procedures for analysing mixture of ink using paper chromatographic technique. 5+5

Q. N. 3 Account for the fact that ‘ the greater the number of alkyl groups , more stable the carbocations and vice versa.’ S_N^2 reaction proceeds with the complete inversion of configuration whereas S_N^1 reaction proceeds with retention plus inversion, discuss with the help of mechanism of the reactions involved? 3+7

Q. N. 4 What is the principal of selecting cathode and anode while constructing Galvanic cell? Describe briefly the various uses of electrochemical series. The standard reduction potential of Cu^{+2}/Cu and Ag^{+}/Ag electrodes are 0.34V and 0.80V respectively. Construct a galvanic cell using these electrodes so that its standard emf is positive 2+5+3

Q. N. 5 Teflon is exceptionally rigid in nature, explain it on the basis of its structure. Manufacture of polythene is an example of addition polymerization, explain on the basis of mechanism involved. Why thinners are used in paint and name any three thinners. 3+4+1+2

Section - B: (1×20=20)

Q. N. 6 a) How do you measure the value of Dissolved Oxygen in water in your Laboratory by Winkler’s method. Write its significance. 10

b) “Human activity is the main cause of increasing greenhouse effect day by day”, explain it. Discuss the nature of stereoisomerism in Tartaric acid 6+4